

PET recovery: diagnosis and improvement campaign

Campaign comparison deck, version 2 (generated from committed results)

PET improvement campaign

PET is diagnostic method development. Nothing here is a publication adoption, and nothing here changes the historical comparison's verdict or thresholds.

The question, and the historical result

The completed matched comparison injected a known truth E_{avail} tilt (amplitude 0.35) into simulated pseudodata and scored the seven-bin recovery of the injected displacement.

arm	mean recovery	adequacy floor	adequate
our production PET	0.3037	0.5560	no
Gregor's pretrained PET2-small	0.4165	0.5560	no

Floor = $0.8 \times$ an acceptance reference 0.6950 built from $1 - (1 - a)^k$ at $k = 3$. Paired ours — theirs = -0.1129 , 95 % interval $[-0.1471, -0.0786]$, $n = 8$ seed pairs; his arm scored better in 8 of 8 pairs.

Verdict (preserved, unchanged by this campaign)

NEITHER_ELIGIBLE / NO_SELECTION. Non-inferiority margin 0.02 and switching margin 0.04 are the historical ones and are used unchanged for every like-for-like statement below.

This campaign asks: what caused the shortfall, what fixes it, whether more events help, whether the adequacy reference is appropriate, and what limits remain.

What actually ran (Phase A audit)

Captured at runtime while the historical code ran unmodified through the guard (Phase A).

1. Gregor's declared optimizer never ran.

Declared: TorchAdamW with weight decay, warmup/cosine schedule, global-norm clipping.
Executed at step 1: Adam from horovod._keras (Horovod-wrapped: yes); weight_decay None, global_clipnorm None, schedule object: no, $\epsilon = 1 \times 10^{-7}$.

The intended unit global-norm clip would have bound on 100 % of updates (pre-clip norms up to 1474).

2. The “identical” step 2 was not identical.

step 2, final stage	theirs	ours
batch size	2048	512
engine steps at gen	586	2344
iteration-0 learning rate	1×10^{-4}	4×10^{-4}

4× more optimizer updates per epoch for ours, and a different validation subset (the engine cuts train/validation in batches).

Defects of fidelity, not established causes

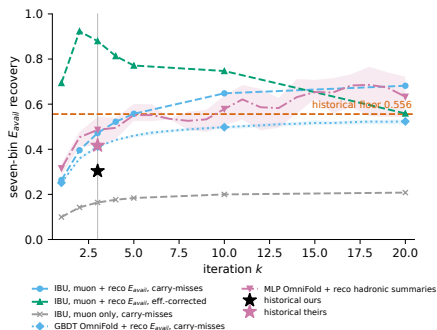
These bias the between-arm comparison; they are not shown to cause the recovery shortfall. The campaign therefore rebuilt the path with explicit per-step recipes; the repaired driver refuses to train when the executed optimizer differs from the declared one.

Where recovery is lost (1): scalar references

At $k = 3$, on the same endpoint (DEV halves; the historical closure reproduced bit-exactly, 31 checks agree: yes):

estimator	recovery
binned IBU, muon kinematics only	0.164
binned IBU, muon + reco E_{avail}	0.472
GBDT OmniFold, muon only	0.140
GBDT OmniFold, + reco E_{avail}	0.412
MLP OmniFold, + all reco hadronic summaries	0.486
historical PET, ours	0.304
historical PET, Gregor's arm	0.417

No response-aware scalar estimator reaches the 0.556 floor at $k = 3$. The shortfall is not PET-specific. Recovery is still climbing: IBU reaches 0.648 at $k = 10$ and 0.683 at $k = 15$; GBDT 0.524 at $k = 20$.

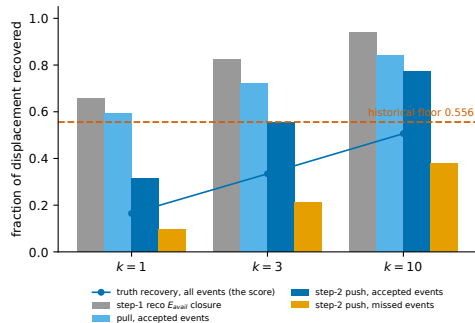


Where recovery is lost (2): step by step inside the unfolding

The truth-side PET is not the bottleneck. Given the known tilt alone it learns it to 0.898 (raw PDG codes), 0.952 (one-hot PDG), 0.973 (+ true E_{avail} , q_3), against 0.33 for any learner restricted to true muon $p_T, p_{\parallel}$ (and 0.999 given true E_{avail}).

Step 1 is not either: the pull closes reco E_{avail} to 0.656 after one iteration, 0.940 by $k = 10$.

The loss is in the misses and in stopping at $k = 3$. Under carry-misses every missed event (58 % of the truth sample) enters step 2 with its previous weight; over accepted events the pull holds 0.591 at $k = 1$, the push keeps 0.313.



k	push	pull	reco E_{av}	acc.	missed
1	0.164	0.191	0.656	0.313	0.098
3	0.334	0.363	0.824	0.554	0.212
10	0.506	0.512	0.940	0.774	0.378

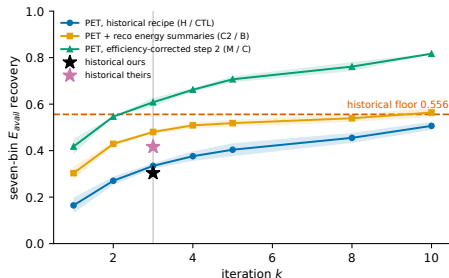
What improves it (1): iterations and the miss rule; the schedule does not

PET through the repaired driver, DEV halves,
 $K = 10$, paired against the historical recipe (2 seeds;
the per-seed paired Δ in brackets):

factor	$k = 3$	$k = 10$	paired Δ , $k = 10$
historical recipe (carry-misses, anneal to 1×10^{-5} , last epoch)	0.334	0.506	—
no forced 1×10^{-5} after iteration 0	0.326	0.496	-0.010 [+0.021,
best-validation epoch handed on	0.290	0.452	-0.054 [-0.047,
efficiency-corrected step 2	0.608	0.817	+0.310 [+0.330, +0.290]

Iterations and the miss rule are the two levers.

Efficiency correction clears the historical floor already at $k = 3$ and lifts the low-acceptance region from 0.17 to 0.67 at $k = 10$. **The schedule does not help:** handing on the best-validation epoch *hurts*.



Bands: min-max over seeds. Efficiency correction is a model-dependence choice; see the robustness slides.

What improves it (2): reconstructed-energy summaries at step 1

Feature arms: carry-misses, one-hot PDG truth side, $K = 10$, 2 seeds, paired against C1 (per-seed Δ in brackets).

arm	$k = 3$	$k = 10$	paired Δ , $k = 3$	paired Δ , $k = 10$
C1 incumbent inputs	0.369	0.525	—	—
C2 + reco energy summaries at step 1	0.480	0.564	+0.112 [+0.106, +0.117]	+0.039 [+0.040, +0.038]
C3 + true E_{avail} , q_3 at step 2	0.384	0.544	+0.015 [+0.011, +0.020]	+0.019 [+0.010, +0.028]
C4 both	0.480	0.558	+0.112 [+0.097, +0.126]	+0.032 [+0.033, +0.032]

- **Explicit reconstructed-energy summaries at the detector step are the one input change that pays:** they speed convergence and lift the accepted events' truth recovery to 0.92 at $k = 10$.
- They do not help the missed events (0.38–0.41 in every arm at $k = 10$), so under the engine's miss rule no feature arm reaches the floor at $k = 3$.
- Truth summaries add little: the truth side already extracts E_{avail} from the cloud.

What improves it (3): miss handling, not the non-iterative form

A bounded scalar benchmark of AUSSIE (arXiv:2602.24282), the non-iterative replacement for OmniFold's pull step. Its first, confounded result was dissected by ablation (DEV populations):

estimator	miss handling	setting	aggregate recovery	seeds
scalar OmniFold (HGB)	carry-misses (the engine's rule)	$k = 50$	0.582	2
scalar OmniFold (HGB)	efficiency-corrected	$k = 50$	0.783	2
AUSSIE	misses pinned to the prior ($\lambda = 1000$)	—	0.648	3
AUSSIE	unconstrained ($\lambda = 0$)	—	0.849	3

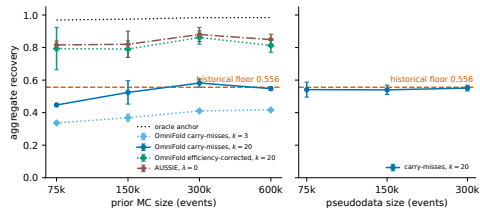
- ▶ **Most of the apparent gain is miss handling:** letting the learned ratio extrapolate to events that fail reconstruction is worth +0.20 (AUSSIE) and +0.20 (OmniFold); AUSSIE keeps a modest edge (+0.07) at matched miss handling.
- ▶ A PET-backbone AUSSIE evaluation is **not justified** on this evidence.
- ▶ Efficiency correction is not free: binned IBU with the acceptance division peaks at 0.923 ($k = 2$) and is back to 0.559 at $k = 20$ with low-acceptance recovery -0.424. It needs a stopping rule and a variance statement.

Does more data help? (Phase D, development stage)

Development-stage evidence (scalar, DEV halves, 222 runs, overlapping draws per size, a per-subset oracle anchor). The confirmatory study on fresh pool-S draws is separate.

axis (aggregate recovery)	75k	150k	300k
OmniFold carry-misses, $k = 3$	0.337	0.369	0.41
OmniFold carry-misses, $k = 20$	0.447	0.524	0.58
OmniFold efficiency-corrected, $k = 20$	0.794	0.791	0.86
AUSSIE, $\lambda = 0$	0.816	0.820	0.88
pseudodata size: carry-misses, $k = 20$	0.541	0.540	0.55
oracle anchor (exact tilt on that subset)	0.969	0.974	0.98

- Prior-MC statistics buy recovery in the historical miss mode (+0.080 at $k = 3$, +0.101 at $k = 20$, 75k→600k); beyond 600k is not measured.
- Pseudodata statistics are not the binding constraint over this range.
- Miss handling outweighs every statistical axis tested (+0.27 at 600k).



Error bars: sd over draws.

The adequacy reference (1): what it measures

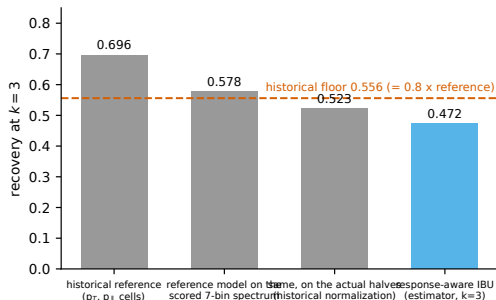
The reference is $1 - (1 - a)^k$ at $k = 3$ with a per-cell acceptance a ; the floor is $0.8 \times$ it.

It is computed on a different quantity from the one scored. It is built on the $(p_T, p_{||})$ cell displacement (0.696); the score is the seven-bin E_{avail} marginal. The model's own assumptions applied to the scored spectrum give 0.578; on the actual halves with the historical pseudodata normalization, 0.523: *below the 0.556 floor derived from it.*

It is not an attainability bound in either direction. Efficiency-corrected IBU exceeds it (0.923 at $k = 2$) and recovers the low-acceptance region (reference 0.014) at 0.76, but only by dividing by acceptance, which the engine's miss rule never does.

The historical normalization assumes the accepted fraction is unchanged by the tilt;

measured ratio 0.8979



Evidence: phase_b/scalar/results/reference_decomposition.json; phase_b/scalar/results/summary.json; ../configuration_comparison/campaign_report.json

The adequacy reference (2): a definitional gap

The gap is definitional, not statistical (fresh pool-S events, 3 replicates per size):

quantity at $k = 3$	historical size	$8\times$ the events
historical reference model (rebuilt on pool S)	0.6950	—
response-aware IBU, $k = 3$	0.472	0.480
reference model realized on the scored object	0.523	0.528

Eight times the statistics closes 0.008 of the 0.223 gap: at least 96 % of it comes from what the reference assumes, not from sample size.

A known-function toy (no smearing): $1 - (1 - a)^k$ at $k = 3$ gives 0.788; the engine's miss rule attains 0.659 (0.025 of the gap from the score's renormalization, 0.104 from the engine's normalization); efficiency correction reaches 0.991, above it. Neither a bound nor a match.

Prospective recommendation only (not adopted; changes no verdict)

A reference must be computed for the estimator's actual miss rule and normalization, on the scored seven-bin spectrum, and a floor derived from it checked on a known-function toy before use. The historical thresholds are retained unchanged for every like-for-like verdict in this campaign.

Robustness (1): what the data can distinguish

A reco-level two-sample classifier at the historical pseudodata size; null from 20 equal-model splits: mean $+0.0002$, 95 % band $[-0.0019, +0.0020]$. Fresh pool-T events.

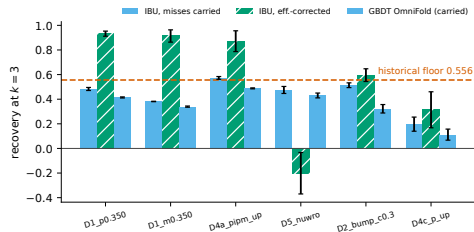
Every predeclared truth distortion is distinguishable (23 of 23), by $2.6\times$ to $62.5\times$ its own ESS-scaled threshold. The weakest is D4d (neutron multiplicity), the hidden-variable test.

The $\pm 5\%$ hadronic scale (R1) is distinguishable. The $\pm 1\%$ muon scale (R2) sits at or below threshold and the cluster smearing (R3) is not distinguishable: **unprobed at this size, not harmless.**

case	$\text{AUC} - \frac{1}{2}$	threshold	disting.
D4d_n_up	+0.0126	0.0023	yes
D4d_n_down	+0.0055	0.0022	yes
R1_x1.05	+0.0218	0.0021	yes
R1_x0.95	+0.0241	0.0021	yes
R2_x1.01	+0.0023	0.0021	yes
R2_x0.99	-0.0002	0.0021	no
R3_s0.10	-0.0014	0.0021	no

Robustness (2): the miss-handling trade-off

distortion ($k = 3$)	IBU carried	IBU eff.-corr.	
D1_p0.350 E_{avail} tilt + (development point)	0.483 ± 0.012	0.932 ± 0.021	0.
D1_m0.350 E_{avail} tilt	0.382	0.913	
D4a_pipm_up $\pi^\pm$ multiplicity up	0.573	0.871	
D5_nuwro NuWro generator reweighting	0.475 ± 0.029	-0.201 ± 0.169	0.
D2_bump_c0.3 bump in E_{avail}	0.514	0.596	
D4c_p_up proton multiplicity up	0.197	0.314	0.112



Mean over 3 replicates on fresh pool-T events ($\pm$ sd where shown). The development-point yardsticks transfer: B1's 0.472 / 0.412 on the historical halves.

Efficiency correction is far ahead when the distortion is a function of the variables the unfold sees, and it fails — moves away from the target — when the distortion changes the event mix inside a truth bin, because it extrapolates an acceptance that no longer holds. Carry-misses is slower but does not fail that way. Neither mode is dominant: the choice is a model-dependence choice and must be stated as one.

Robustness (3): hidden variable and response scale

Hidden variable (D4d: neutron multiplicity up/down, invisible in E_{avail} , visible to the detector). Every estimator ends farther from the target than the untouched prior: the projection on the injected direction is negative (the wrong way) in 14 of 18 estimator $\times$ replicate cells at $k = 3$ (development tilt: positive in 9 of 9). This is the hidden-variable limit any unfolding in these observables carries. That efficiency correction fails *specifically* worse here is **not established** at this sample size.

Detector response. A $\pm 5\%$ hadronic energy-scale error (R1) combined with the development tilt:

estimator ($k = 3$)	same response	R1 $\times 1.05$ (Δ)	R1 $\times 0.95$ (Δ)
IBU, misses carried	0.483	0.546 (+0.063)	0.418 (−0.065)
GBDT OmniFold	0.414	0.476 (+0.061)	0.356 (−0.058)

Shifts average 0.06 in magnitude, $3.1\times$ the historical 0.02 non-inferiority margin. Same-response closure cannot see this: **any candidate's recovery statement is conditional on the simulated response.**

The confirmatory result: pending

Confirmatory results pending

The FINAL comparison (fresh pool F, n independent replicates) has not been harvested into `confirm_results.json`. No confirmatory decision, superiority, non-inferiority or adequacy statement is made in this deck.

Frozen before any fresh-pool row was read (protocol amendment 2): **CTL** = historical recipe at $k = 3$; **A** = the same run at $K^* = 10$ (iterations alone); **B** = C2 inputs at K^* (carry-misses); **C** = efficiency-corrected step 2 at K^* . Adequacy judged like-for-like at $k = 3$ and at K^* against the unchanged historical floors.

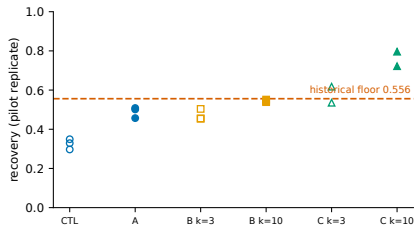
PILOT, pool P	P0	P1	P2
CTL (hist. recipe, $k = 3$)	0.297	0.329	0.349
A (same run, $K^* = 10$)	0.457	0.508	0.501
B (C2 inputs), $k = 3$	0.454	0.455	0.504
B (C2 inputs), $K^* = 10$	0.542	0.551	0.539
C (eff.-corr.), $k = 3$	0.535	0.617	—
C (eff.-corr.), $K^* = 10$	0.722	0.796	—
oracle anchor	0.987	0.985	0.982

FINAL sized from these at $n = 12$ (uncapped 200; capped by pool

F).

PILOT observations are not confirmatory. They size FINAL and are reported separately; they never enter the FINAL

interval, and no decision is drawn from them.



What a terminal result cannot authorize

A terminal result of this campaign cannot authorize

- ▶ adoption of any estimator for publication;
- ▶ a change to the historical comparison's verdict (NEITHER_ELIGIBLE / NO_SELECTION, preserved) or to its thresholds;
- ▶ C_{stat} / C_{ML} or any uncertainty product, any systematic, any central-value change, any Gate-6 action;
- ▶ any real-data unfolding.

Recalibrated references and recommended configurations are **prospective recommendations only**. Every result in this deck is from simulation: pseudodata built from signal Monte Carlo with a known injected truth.

Scope. PET is diagnostic method development. Nothing here is a publication adoption, and nothing here changes the historical comparison's verdict or thresholds.

Limitations and the next choice

Limitations

- ▶ DEV-stage PET results rest on 2 seeds per arm. The repaired driver reproduces the historical recipe (0.311 over 4 seeds, sd 0.03); effects below about that spread are reported as unresolved, not as nulls.
- ▶ Recovery statements are conditional on the simulated detector response (R1 moves them by more than the margins) and cannot see hidden variables (D4d).
- ▶ Efficiency correction needs a stopping rule and a variance statement, not only a mean recovery.
- ▶ Coverage has not been run: protocol §7 reserves it for a candidate adequate or best-in-class on FINAL.

Next choice

- ▶ Confirmatory status: the FINAL comparison and the PET stress set are pending; until they land, every candidate statement above is development-stage evidence on the DEV halves.
- ▶ The miss rule is a model-dependence choice (carry-misses: slower, does not move away; efficiency correction: faster, fails when the event mix inside a truth bin changes). Whichever is recommended must be stated as that choice.
- ▶ A reference computed for the estimator's own miss rule and normalization, on the scored spectrum, is a prospective recommendation to be checked on a known-function toy before use.

PET is diagnostic method development. Nothing here is a publication adoption, and nothing here changes the historical comparison's verdict or thresholds.

Appendix

Backup

Appendix: other executed facts (Phase A)

- ▶ Every fit after iteration 0 ran at a forced learning rate (1×10^{-5}), so the tuned rate governed iteration 0 only.
- ▶ Early stopping was inert against 8 epochs: each fit handed on its **last-epoch** weights, while the validation-loss minimum was the last epoch in only 7 of 48 final-stage fits for his arm and 30 of 48 for ours.
- ▶ Step 1 normalizes each class to a fixed total, discarding the physical class ratio.
- ▶ The step-2 truth cloud feeds **raw PDG codes** as a continuous column into Dense layers and the k -NN features.
- ▶ **Not defective:** his pretrained weights survive cloning exactly (176 of 176 tensors at the first optimizer step); his token features are the converted $\eta/\phi/\log p_T/\log E$ set.

Appendix: Phase D, all prior-size curves

Prior-MC size axis, mean $\pm$ sd over draws (Phase D, development stage):

estimator	75k	150k	300k	600k
OmniFold eff.-corrected, $k = 3$	0.749 ± 0.043	0.748 ± 0.032	0.827 ± 0.021	0.793 ± 0.025
AUSSIE, $\lambda = 1000$ (misses pinned)	0.679 ± 0.201	0.641 ± 0.083	0.663 ± 0.027	0.648 ± 0.031
OmniFold carry-misses, $k = 3$	0.337 ± 0.008	0.369 ± 0.022	0.411 ± 0.004	0.416 ± 0.009
OmniFold carry-misses, $k = 20$	0.447 ± 0.007	0.524 ± 0.072	0.582 ± 0.025	0.548 ± 0.012
OmniFold eff.-corrected, $k = 20$	0.794 ± 0.130	0.791 ± 0.051	0.862 ± 0.041	0.813 ± 0.042
AUSSIE, $\lambda = 0$	0.816 ± 0.024	0.820 ± 0.081	0.880 ± 0.043	0.849 ± 0.033

The efficiency-corrected and AUSSIE curves are flat in prior size: the signature of an estimator limited by its extrapolation assumption rather than by statistics.

Appendix: Phase E scalar yardsticks, all predeclared cases

Predeclared truth distortions and R1 combined with the development tilt; mean recovery over 3 replicates, fresh pool-T events.

case	IBU carried $k=3$	$k=10$	IBU eff. $k=3$	$k=10$	GBDT $k=3$	$k=10$
D1_m0.175	0.38	0.53	0.92	0.62	0.28	0.36
D1_m0.350	0.38	0.52	0.91	0.78	0.34	0.43
D1_m0.700	0.39	0.51	0.88	0.88	0.36	0.46
D1_p0.175	0.45	0.62	0.88	0.79	0.34	0.41
D1_p0.350	0.48	0.66	0.93	0.86	0.41	0.50
D1_p0.700	0.56	0.74	0.96	0.86	0.51	0.60
D2_bump_c0.3	0.51	0.68	0.60	0.51	0.32	0.37
D2_bump_c1.0	0.35	0.51	0.54	0.52	0.22	0.30
D3_m0.35	0.34	0.44	0.71	0.72	0.26	0.31
D3_p0.35	0.40	0.51	0.68	0.58	0.32	0.37
D4a_pipm_down	0.50	0.63	0.87	0.66	0.40	0.47
D4a_pipm_up	0.57	0.71	0.87	0.78	0.49	0.56
D4b_pi0_down	0.51	0.67	0.84	0.54	0.38	0.47
D4b_pi0_up	0.56	0.74	0.92	0.76	0.46	0.54
D4c_p_down	0.29	0.41	0.26	-0.40	0.20	0.20
D4c_p_up	0.20	0.19	0.31	-0.40	0.11	0.13
D4d_n_down	-0.41	-0.90	-1.95	-4.91	-0.06	-0.09
D4d_n_up	-0.32	-0.59	-0.46	-2.12	-0.15	-0.15
D5_fsi_frabs	0.61	0.82	0.86	0.67	0.55	0.71
D5_fsi_frinel	0.61	0.81	0.86	0.67	0.56	0.71
D5_genie_mec	0.62	0.80	0.84	0.65	0.57	0.72
D5_gibuu	0.55	0.69	0.65	0.50	0.44	0.50
D5_nuwro	0.48	0.65	-0.20	-0.00	0.43	0.50
R1_x0.95+D1_p0.350	0.42	0.56	0.80	0.85	0.36	0.42
R1_x1.05+D1_p0.350	0.55	0.76	0.87	0.69	0.48	0.59

Appendix: evidence files

Every number in this deck is read from these committed files by `slides/make_campaign_deck.py`; `slides/deck_numbers.json` records the field (or derivation) behind each, and `slides/CLAIM_INDEX-deck.md` maps every claim to them. Paths relative to `nd-unfolding/pet/improvement_campaign/`; sha256 prefixes.

file	sha256
../configuration_comparison/campaign_report.json	96955a10222b6c95
confirm/results/confirm_results.json	4a00848e37a08f67
phase_a/receipts/historical_receipts.json	938b97521cc77e82
phase_a/receipts/runtime_audit_theirs.json	8dadb3f32d3fec72
phase_b/pet/results/summary.json	ed81c51f67e9df24
phase_b/scalar/results/reference_decomposition.json	aa831560259f1255
phase_b/scalar/results/summary.json	50665f198fa4ad73
phase_d/results/scalar_scaling.json	a0eadf592dc8e42a
phase_e/results/identifiability.json	856aa31aae209fe8
phase_e/results/reference_assessment.json	610e643515efbc34
phase_e/results/references.json	5c27d2ce21097102
phase_e/results/toy_reference.json	dd5d85c699adb3d
phase_f/results/aussie_ablation.json	cbe81805d792468e

Plus 16 per-run PET score files under `phase_b/pet/results/` (hashes in `deck_numbers.json`).